Wong Yeow Kang

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PROFILE SUMMARY

Intellectually curious and technically driven AI-focused engineer with a strong foundation in programming, data analytics and machine learning. Thrives in fast-paced, analytical environments, with a particular interest in **applied AI, reinforcement learning and decision-making systems**. Motivated by developing scalable, data-driven solutions that translate research ideas into real-world impact. Seeking roles in **AI/ML, Data or Software Engineering**.

EDUCATION

NATIONAL UNIVERSITY OF SINGAPORE Aug 2023 - May 2027

Bachelor of Science in Business Analytics (Honors), Machine Learning Specialisation

- Grade Point Average: 4.47/5.0
- Activities: **NUS Overseas Colleges (Silicon Valley)**, Google Developer Student Club, NUS FinTech Society
- Relevant Coursework: Statistics, Linear Algebra for Computing, Advanced Calculus, Data Analytics and Visualization, Programming Methodology (Python and Java), Data Structures and Algorithms, Feature Engineering for Machine Learning, Application Systems Development

WORK EXPERIENCE

WORKATO HQ PALO ALTO *Partner Solutions Consultant (AI Apps)* Aug 2025 - Present

- Deployed a **workflow automation app** enabling Workato partners to instantly retrieve company-wide certification progress, reducing turnaround time from 3 days (manual Jira requests) to 5 seconds (automated report.
- Built AI demo solutions integrating Workato's **Model Context Protocol (MCP)** servers with LLM clients (e.g ChatGPT) and local LLM runtimes (e.g LMStudio), enabling partners to expose workflows designed within the Workato platform as tools to the MCP server.
- Developed **LangChain** pipelines with Workato MCP servers and vector databases such as ChromaDB for **retrieval-augmented generation (RAG)**, demonstrating scalable context injection for enterprise use cases.

MARIBANK *Data Analyst Intern (Consumer & SME)* Jan 2025 - Apr 2025

- Analyzed month-over-month changes in SMS notification engagements leveraging MySQL, providing data-driven insights, **reducing operational costs by 73%** from December'24 to January'25 for the bank.
- Built frontend tracking system by creating user tag attributes for different banking customers to **monitor user behavior** and identify drop-off points in the customer onboarding process, improving banking onboarding rate.
- **Initiated the automation** of monthly analysis and email notification delegation using Google Apps Script (JavaScript), eliminating the need for manual data updates, and **reducing irrelevant email spam by over 40%**.

NUS COMPUTING *Teaching Assistant (CS1010A Programming Methodology)* Aug 2024 - Dec 2024

- Taught foundational programming concepts in **Python**, including **Data Structures and Algorithms** and **OOP**.
- Led weekly tutorial sessions for a class of 12 students, providing hands on guidance and conducting evaluation of assignments across **practical coding exercises** and **theoretical quizzes**.
- Achieved an **80% distinction rate** by delivering targeted support and engaging students in revision exercises.

PROJECTS

Trading Project (GitHub) Dec 2025 - Present

A keyword-specialized reinforcement learning with verifiable rewards (RLVR) system for news-driven market prediction.

- Building a trading signal pipeline that converts 30-day rolling financial news into structured sentiment features using a shared **FinBERT** backbone, framing market prediction as a contextual bandit problem.
- Designing keyword-specific **reinforcement learning policies trained with verified market rewards**, enabling adaptive next-day prediction without human feedback.
- Implemented a modular architecture supporting parallel asset specialization (e.g Gold, NVIDIA) with shared representations and independently learned decision heads.

Forged Test Portal, Freelance Web Development (Website Link) Nov 2025 - Present

A full-stack athlete testing & reporting platform with secure Auth and automated report generation.

- Built a production-ready client-server web application for athlete testing using React + Vite + TypeScript, featuring Supabase-based authentication, role-based access control and protected routes for client and staff.
- Implemented reusable performance metric calculations (e.g lactate thresholds) in a shared library consumed by both frontend and backend, and automated HTML to PDF report generation for deployment ready workflows.

SpotiFind, Music Recommendation System (GitHub) Jan 2025 - May 2025

A machine learning-based music recommender using audio features, lyric sentiment, and hybrid filtering.

- Developed a **machine learning-based recommendation system** leveraging acoustic features, harmonic pattern analysis and BERT-based sentiment analysis for contextual relevance in lyrics.
- Engineered audio-based data for over 250,000 songs, applying K-means clustering to uncover genre groupings.
- Applied **collaborative filtering** and **content-based models**, analyzing spectrogram data with cosine similarity to recommend tracks with similar frequency distributions.

CO-CURRICULAR ACTIVITIES

NUS FINTECH SOCIETY, Machine Learning Analyst Trainee Aug 2024 - Dec 2024

- Utilized Python libraries such as Pandas and Seaborn for data analytics and visualisation.
- Gained hands-on experience with data preprocessing, feature engineering, and building machine learning pipelines using Scikit-learn through labs.

GOOGLE DEVELOPER STUDENT CLUB (NUS), Business Associate Aug 2023 - May 2024

- Collaborated with a multidisciplinary team of developers and UI/UX designers to create Project Ignite, leveraging technology to streamline volunteer recruitment and management processes for non-profit startups, driving social impact and community engagement.

TECHNICAL SKILLS AND PROFESSIONAL CERTIFICATIONS

- Written & Spoken Languages: English, Mandarin
- Programming Languages: Python, Java, R, SQL, JavaScript, SQL
- Design and Front-End: Vue.js, React.js, Next.js, HTML, CSS, Tailwind, Figma, Canva
- Back-End Web Development: Flask, Node.js, Express.js
- AI and Machine Learning: PyTorch, scikit-learn, NumPy, HuggingFace Transformers, LangChain
- Professional Certifications: CS50AI Introduction to AI with Python, Google Advanced Data Analytics Certification, Quantitative Risk Management in Python, Associate AI Engineer for Developers (DataCamp)